



CEPIITD

Online PG Diploma: Advanced Communication Engineering with Quantum and AI Integration

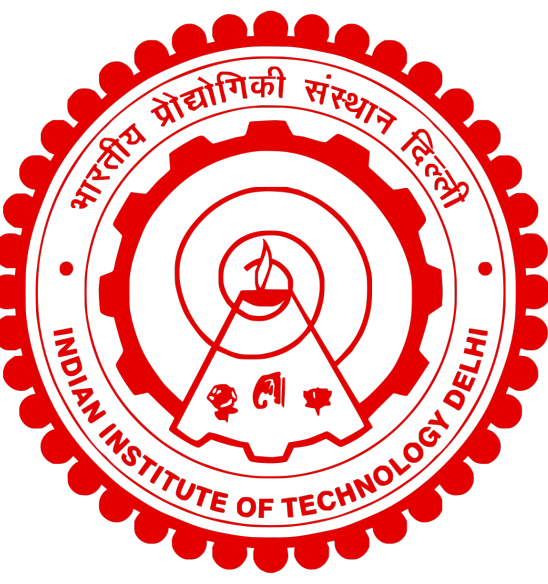
Course: Quantum Information and Security

Instructor: Dr. Neel Kanth Kundu

Assistant Professor

Centre for Applied Research in Electronics (CARE)
Bharti School of Telecom Tech. & Mgmt. (Joint Faculty)
Indian Institute of Technology Delhi

Course Contents



- Classical Information Theory
 - Shannon's Source and Channel Coding Theorems
- Quantum Information Theory
 - Holevo Capacity
- Quantum Key Distribution
 - DV-QKD and CV-QKD Secrecy Analysis



Quantum Information and Security

Module-1

Classical Information Theory

Instructor: Dr. Neel Kanth Kundu

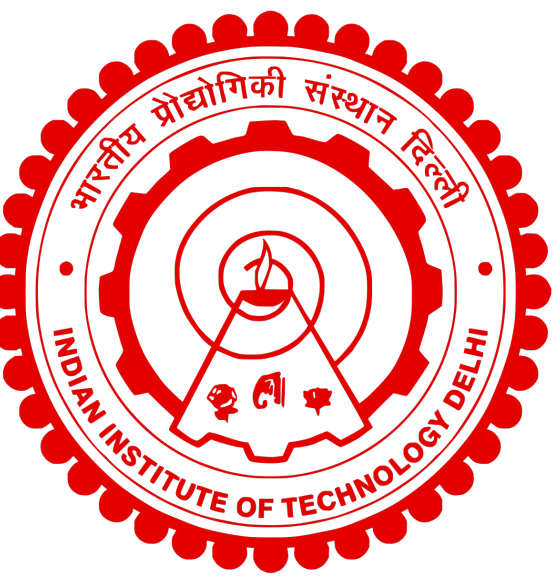
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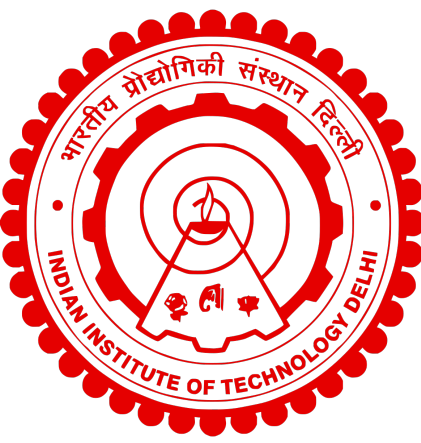
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Overview

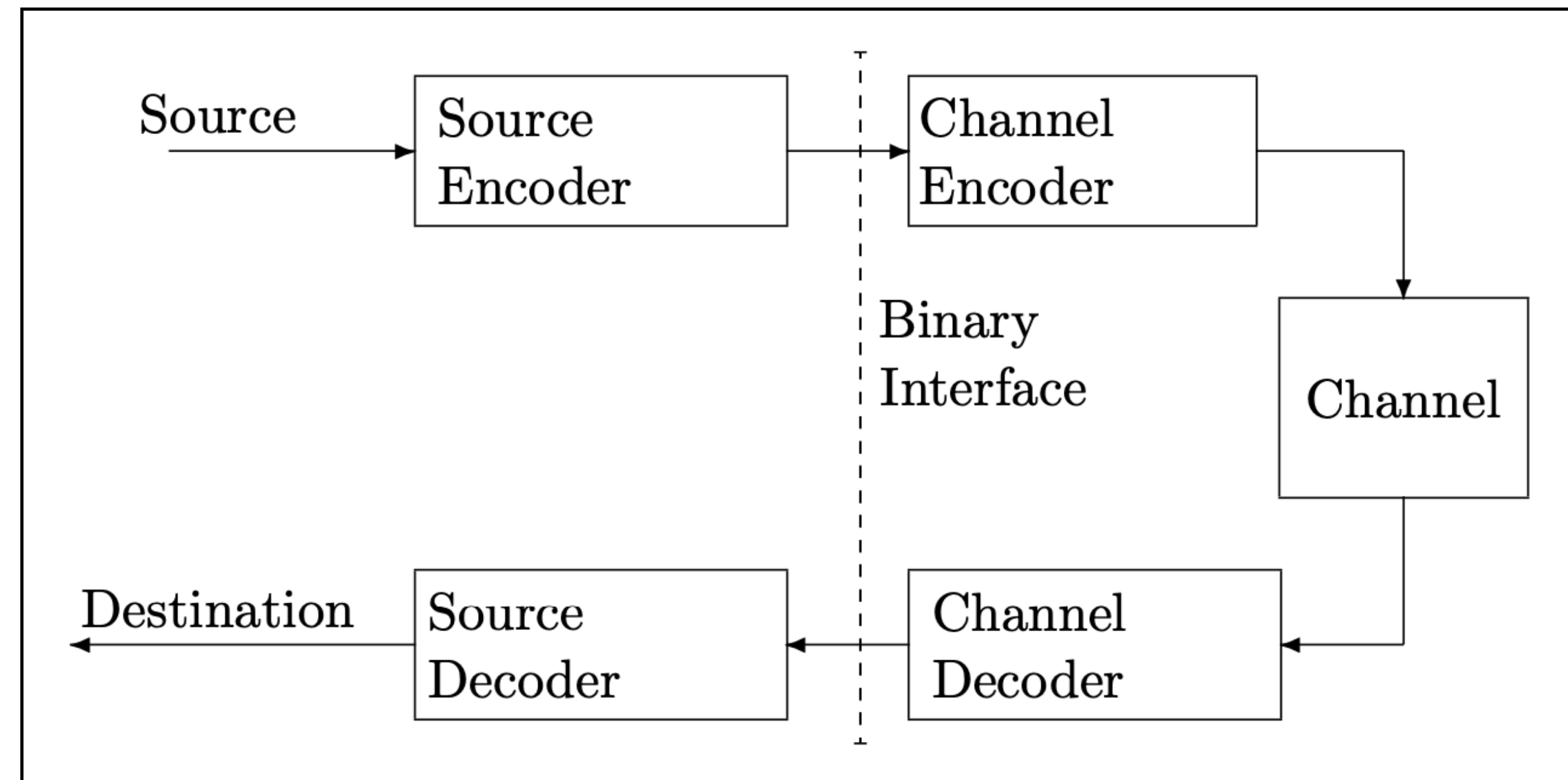


- Motivation
- Classical Entropy
- Noiseless Source Coding: Data Compression
- Noisy Channel Coding: Channel Capacity

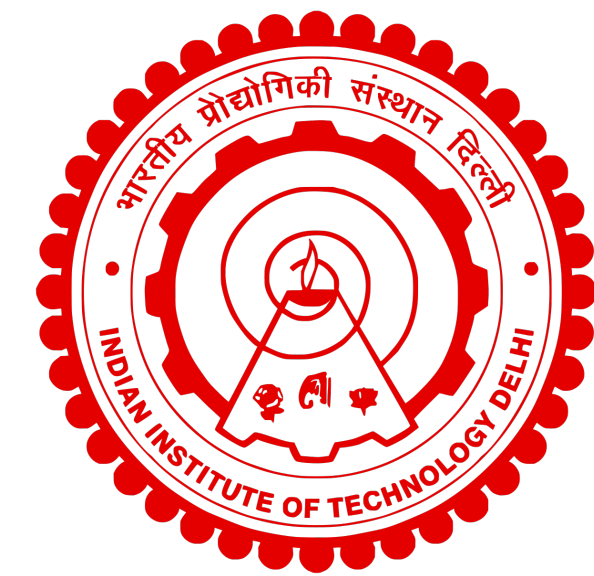
Motivation



- Goals of Information theory :
 1. Define *information* mathematically and quantitatively
 2. Represent information in efficient way (data compression) for storage and transmission
 3. Ensure the protection of information (through encoding) in presence of noise & impairments
- Classical Information Theory (CIT): Achieve above goals in accordance with laws of classical physics
- Quantum Information Theory (QIT): Achieve above goals in accordance with quantum mechanical principles
- QIT is richer than CIT due to intriguing resources like entanglement

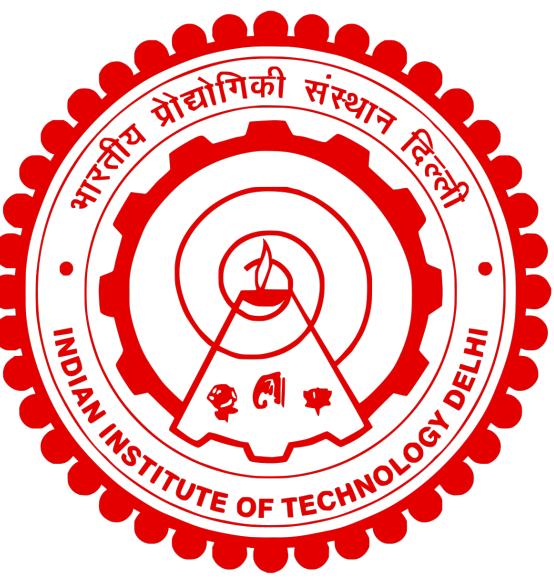


Classical Information Theory



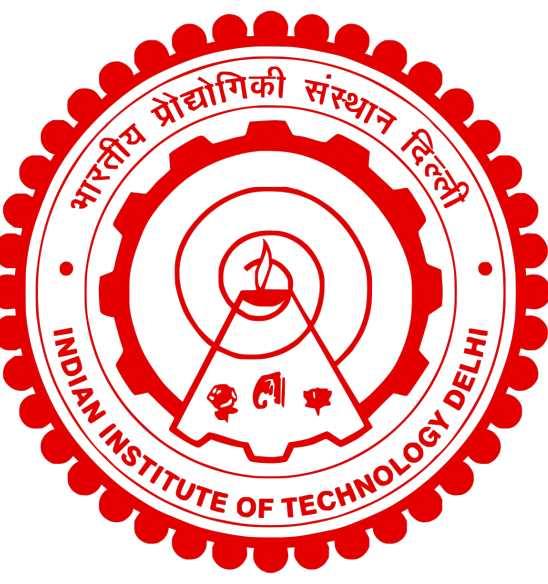
- **Entropy** quantifies information and uncertainty.
- Important for **security proofs** in Quantum Key Distribution (QKD).
- Helps estimate adversary's knowledge in QKD.
- Understanding classical entropy builds intuition for quantum cases.
- **Claude Shannon** (1948) laid foundations of classical information theory.
- **Key Questions:**
 1. How many bits are needed to compress data reliably?
 2. How much information can be reliably transmitted through a channel?

Overview

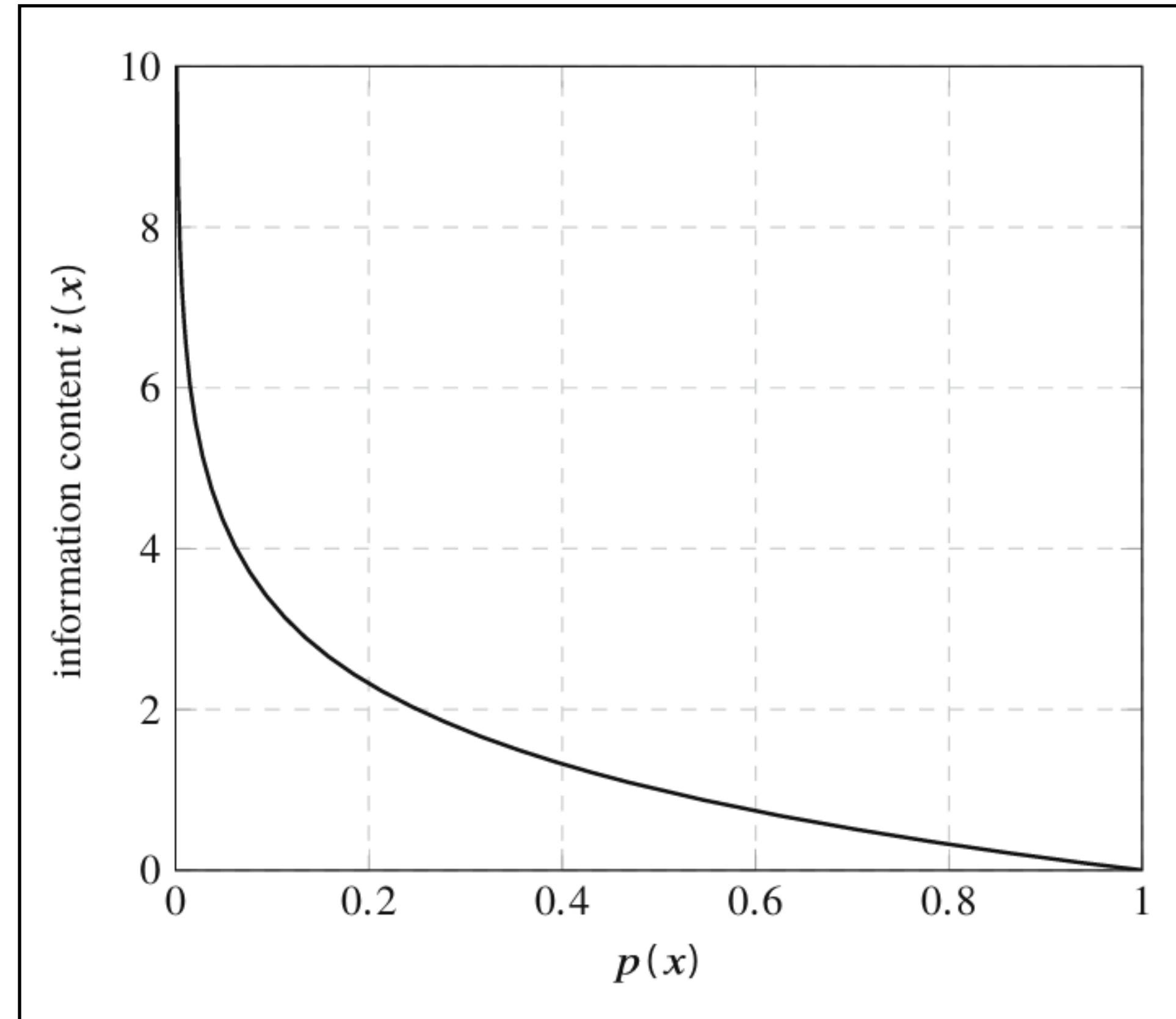


- Introduction
- **Classical Entropy**
- Noiseless Source Coding: Data Compression
- Noisy Channel Coding: Channel Capacity

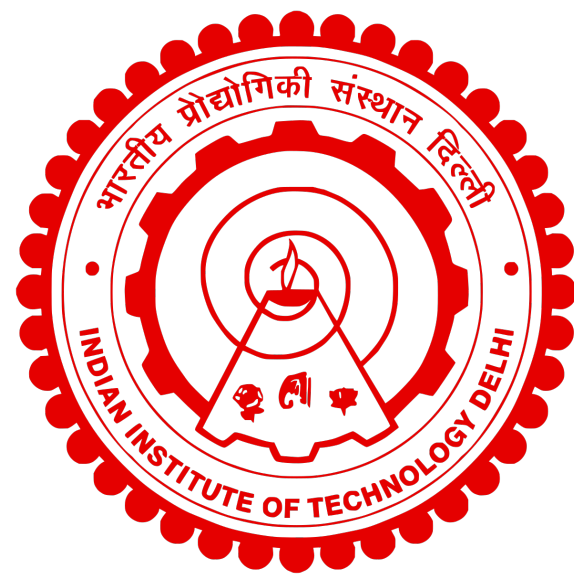
Quantifying Information



- Let X be a random variable with realizations , $x \in \mathcal{X}$ and probability $p_X(x)$
- Information content (surprise) of outcome x : $i(x) = -\log_2(p_X(x))$
- Units of information: **bits**



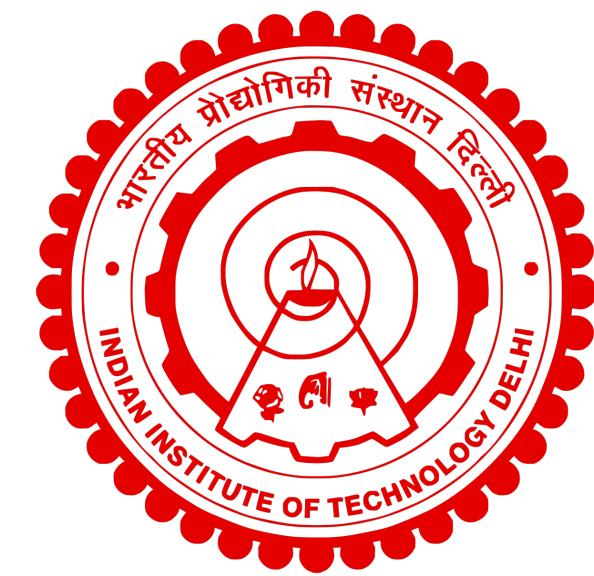
Properties of Information Content



- **Label-Invariance:**
 - Information content depends only on the probability of the outcome, not its label.
 - Example: Variables with values $\{0,1\}$ or $\{+, -\}$ and same probabilities carry the same information.
- **Continuity:** The function $i(x)$ is continuous for all $p_X(x) \in (0,1]$.
- **Monotonicity and Non-negativity:**
 - Unlikely events (small p) produce higher information content.
 - Since $\log_2(p) \leq 0$, the information content $i(x) \geq 0$ for all x .
- **Additivity for Independent Events:** If two random variables X_1 and X_2 are independent, then the total information content satisfies: $i(x_1, x_2) = i(x_1) + i(x_2)$
- When outcomes are independent, the total information is simply the sum of individual surprises.
 - **Proof:**

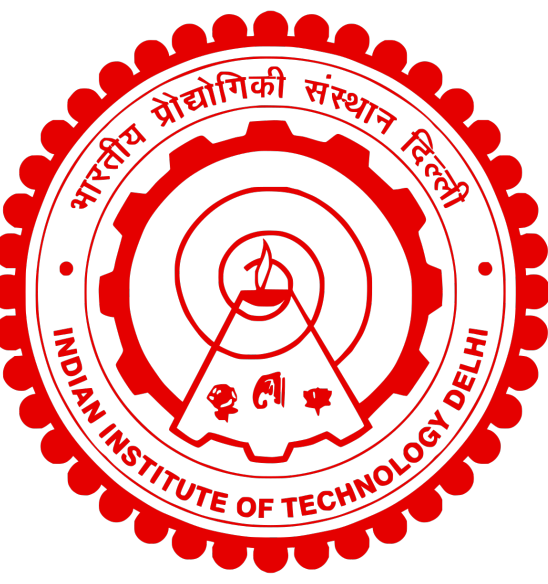
$$\begin{aligned} i(x_1, x_2) &= -\log_2(p(x_1, x_2)) \\ &= -\log_2(p(x_1) \cdot p(x_2)) \quad (\text{independence}) \\ &= -\log_2(p(x_1)) - \log_2(p(x_2)) \\ &= i(x_1) + i(x_2) \end{aligned}$$

Shannon's First Question: Compression



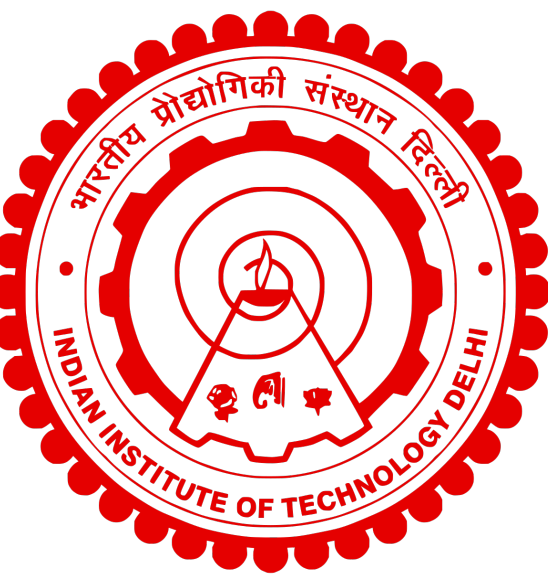
- Q1: How many bits are required to reliably compress a given amount of information?
- A1: According to **Shannon's Noiseless Coding Theorem**, the optimal average number of bits required to encode symbols from a source with random variable X is given by its entropy.
- **Entropy:** $H(X) = - \sum_x p(x) \log p(x)$
- **Explanation:**
 - Entropy $H(X)$ measures the average uncertainty in a random variable.
 - It gives the lower bound for lossless compression — no method can, on average, compress below this limit without losing information.
 - For large enough sequences, it's possible to compress to approximately $H(X)$ bits per symbol.

Shannon Entropy

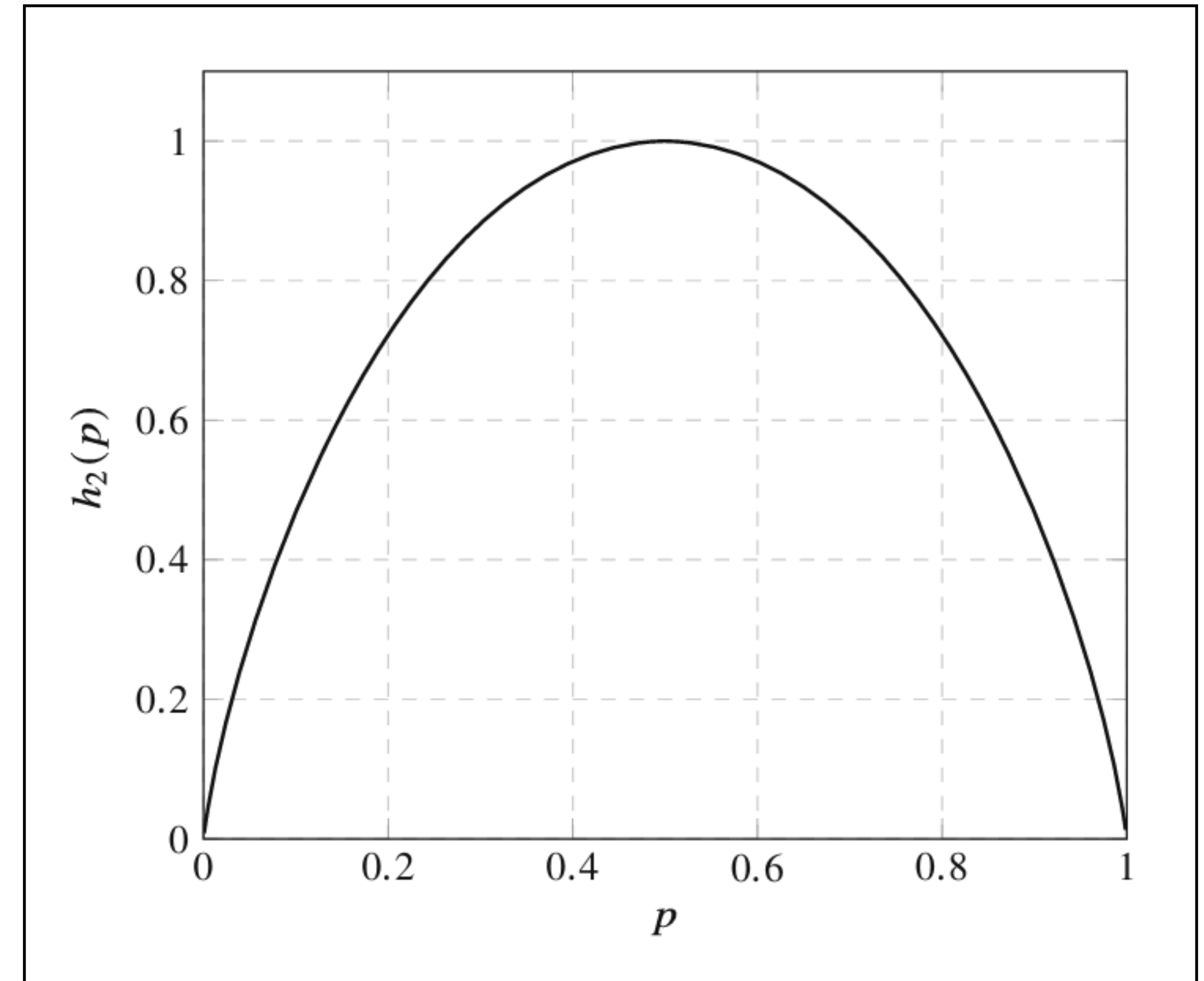


- **Goal:** Measure the average information content of a random variable X .
- **Definition:** The entropy of a discrete random variable X with probability distribution $p_X(x)$ is:
$$H(X) = - \sum_x p_X(x) \log_2 p_X(x)$$
 - Convention: $0 \cdot \log(0) = 0$
 - Justification: $\lim_{p \rightarrow 0} p \log p = 0$
- **Example-1:** Fair Coin Toss
 - Let X be the outcome of a fair coin: $p(H) = p(T) = \frac{1}{2}$.
 - Then $H(X) = - \left(\frac{1}{2} \log_2 \frac{1}{2} + \frac{1}{2} \log_2 \frac{1}{2} \right) = 1$ bit
 - **Interpretation:** Each outcome gives us 1 bit of information.
- **Example-2:** A fair 8-sided die: $P(i) = \frac{1}{8}, i = 1, \dots, 8$
 - Entropy: $H(X) = - \sum_{i=1}^8 \frac{1}{8} \log_2 \left(\frac{1}{8} \right) = \log_2(8) = 3$ bits

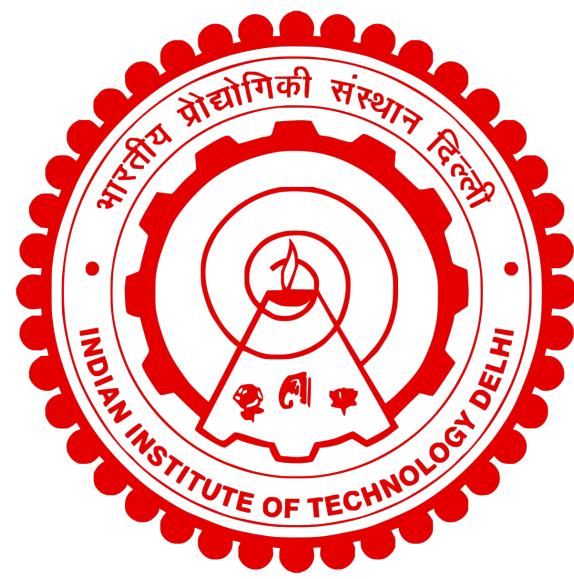
Binary Entropy



- Let X be a random variable with values in $\{0,1\}$.
- Probability distribution:
 $P(X = 0) = p, P(X = 1) = 1 - p$
- Binary Entropy is defined as:
 $h_2(p) = -p \log p - (1 - p) \log(1 - p)$
- Symmetry: $h_2(p) = h_2(1 - p)$
- Maximum entropy at $p = 0.5$ (unbiased coin)
- Minimum entropy ($h_2(p) = 0$) when $p = 0$ or 1 (deterministic)



Shanon Entropy Properties - I



1. **Entropy is Non-negative:** $H(X) = - \sum_x p_X(x) \log p_X(x) = \sum_x p_X(x) \cdot i(x) \geq 0$

- Entropy represents expected information.
- Since $i(x) = -\log p_X(x) \geq 0$, we have $H(X) \geq 0$.

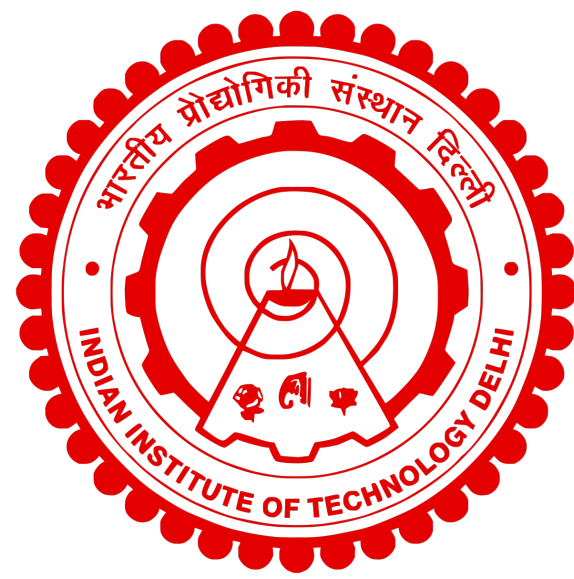
2. **Entropy is Concave:** $H(\lambda p_1 + (1 - \lambda)p_2) \geq \lambda H(p_1) + (1 - \lambda)H(p_2)$

- Entropy increases under the averaging of distributions.
- This reflects increased uncertainty when mixing sources.

3. **Invariance Under Relabeling:**

- Entropy depends only on the probabilities, not on the labels of outcomes.
- Permuting or renaming outcomes does not affect $H(X)$.

Shannon Entropy Properties - II



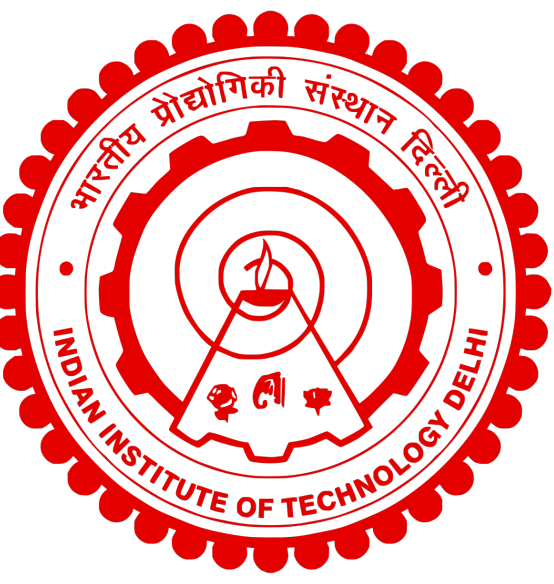
4. **Zero Entropy for Deterministic Variables:** $H(X) = 0 \iff p_X(x) \in \{0,1\}$ for all x

- If all probability mass is on one value, no uncertainty exists.
- Learning X yields no new information.
- **Proof:** $H(X) = 0 \implies p_X(x)\log(p_X(x)) = 0$ for every $x \in \mathcal{X}$.
- Two possibilities: $p_X(x) = 0$ or $p_X(x) = 1$. Because $\sum_x p_X(x) = 1$, there is exactly one x_0 for which $p_X(x_0) = 1$.

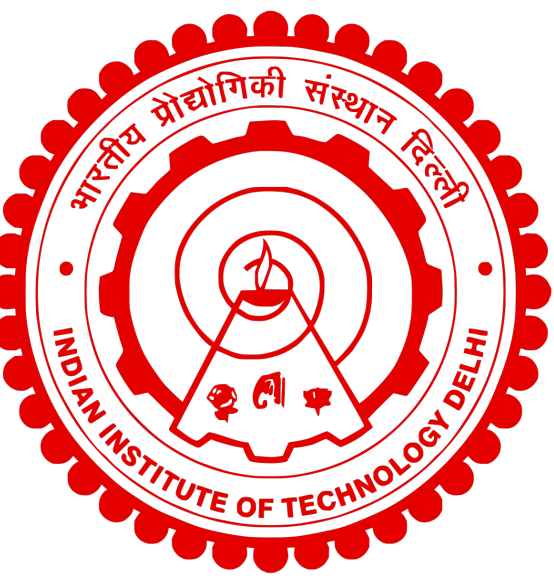
5. **Maximum Value of Entropy:** $H(X) \leq \log |\mathcal{X}|$

- where $|\mathcal{X}|$ is the cardinality of the alphabet
- Equality if all outcomes are equally likely, that means X is a uniform random variable.

Overview



- Introduction
- Classical Entropy
- **Noiseless Source Coding: Data Compression**
- Noisy Channel Coding: Channel Capacity



Example: Fixed-Length Coding

- **Source:** $X \in \{A, B, C, D\}$ with probabilities:
 $P(A) = \frac{1}{2}, \quad P(B) = \frac{1}{4}, \quad P(C) = \frac{1}{8}, \quad P(D) = \frac{1}{8}$
- **Fixed-length coding:** Use 2 bits per symbol: $A \rightarrow 00, \quad B \rightarrow 01, \quad C \rightarrow 10, \quad D \rightarrow 11$
- **Expected codeword length:** 2 bits
- **Shannon Entropy:**

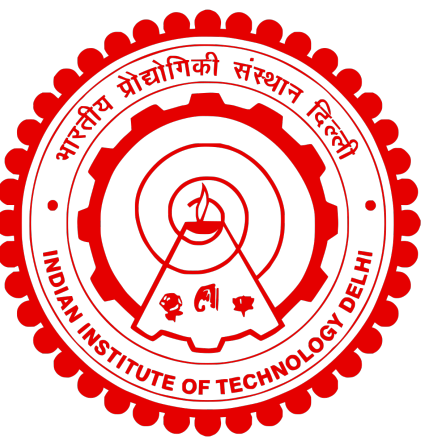
$$\begin{aligned} H(X) &= - \left(\frac{1}{2} \log_2 \frac{1}{2} + \frac{1}{4} \log_2 \frac{1}{4} + 2 \cdot \frac{1}{8} \log_2 \frac{1}{8} \right) \\ &= \frac{1}{2} + \frac{1}{2} + \frac{3}{4} = \frac{7}{4} = 1.75 \text{ bits} \end{aligned}$$

- *Entropy is lower than the expected length of fixed-length coding.*

Example: Variable-Length Coding

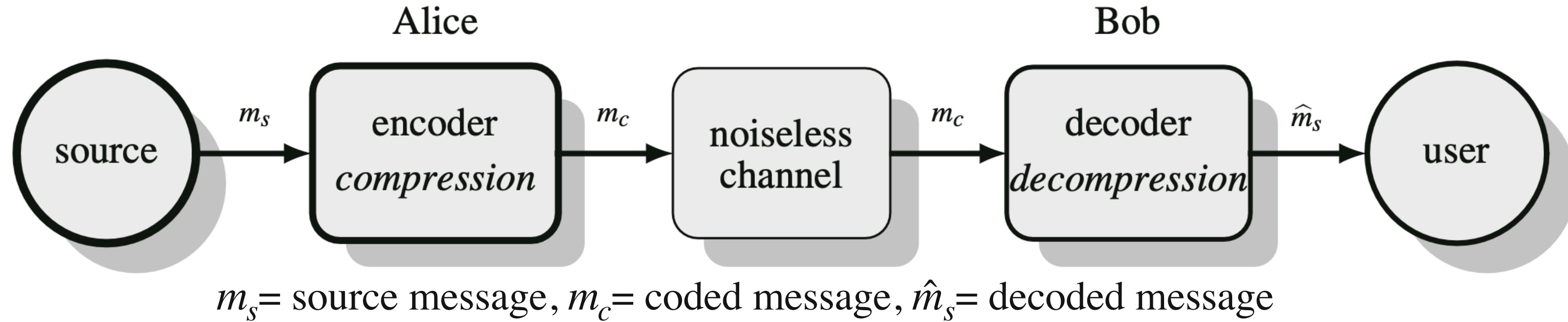


- **Variable length coding:** $A \rightarrow 0$, $B \rightarrow 10$, $C \rightarrow 110$, $D \rightarrow 111$
- **Expected codeword length:** $\frac{1}{2} \cdot 1 + \frac{1}{4} \cdot 2 + \frac{1}{8} \cdot 3 + \frac{1}{8} \cdot 3 = \frac{7}{4} = 1.75$ bits
- **Conclusion:**
 - This variable-length code achieves the Shannon limit.
 - Efficient compression: matches $H(X)$.
 - The code is prefix-free implies uniquely decodable.
 - A coding scheme must ensure unique decodability.



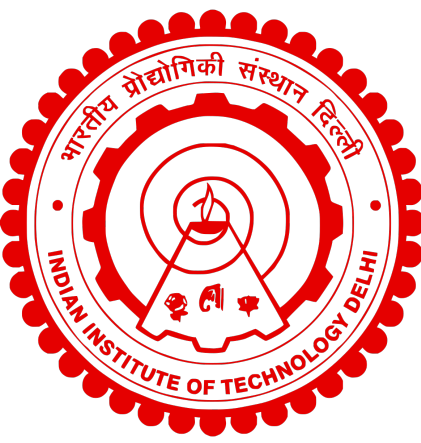
Noiseless Source Coding: Data Compression

- Efficient representation of messages produced by a source of information with redundancy
- Source: a sequence of IID random variables
- $A \sim (\mathcal{A}, p_A)$, where $\mathcal{A} = \{1, \dots, K\}$, word: $A^L := (A_1, \dots, A_L)$, realisations: $a^L := (a_1, \dots, a_L) \in \mathcal{A}^L$ with probability: $P[A^L = a^L] = p_A(a_1) \cdots p_A(a_L)$



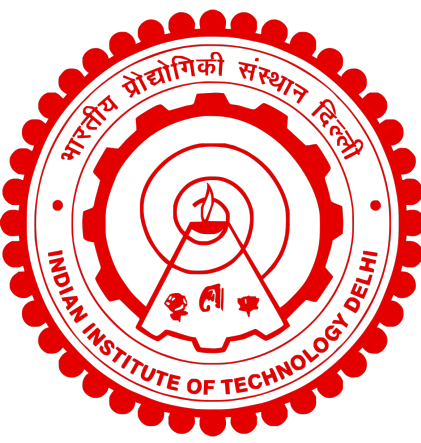
- **Theorem:** (Law of large numbers, LLN) In an IID source, the mean estimator: $\mathcal{E}(A^L) := \frac{1}{L} \sum_{n=1}^L A_n$, converges to the mean $m(A) = E[A]$: $\lim_{L \rightarrow \infty} \mathcal{E}(A^L) = m(A)$ (in probability) i.e., for any $\varepsilon > 0$, $P[|\mathcal{E}(A^L) - m(A)| > \varepsilon] \rightarrow 0$ as $L \rightarrow \infty$.
- Entropic Application of LLN: Information of a source word A^L : $i(A^L) = -\log[p_A(A_1) \cdots p_A(A_L)]$ s.t. entropy: $H(A^L) = LH(A)$ and entropy estimator: $\mathcal{J}(A^L) := \frac{1}{L} i(A_1, \dots, A_L) = \frac{1}{L} \sum_{n=1}^L i(A_n) = -\frac{1}{L} \sum_{n=1}^L \log p_A(A_n)$
- **Theorem:** In an IID source, the entropy estimator $\mathcal{J}(A^L)$ converges to entropy: $\lim_{L \rightarrow \infty} \mathcal{J}(A^L) = H(A)$ (in probability)

Shannon's Source Coding Protocol

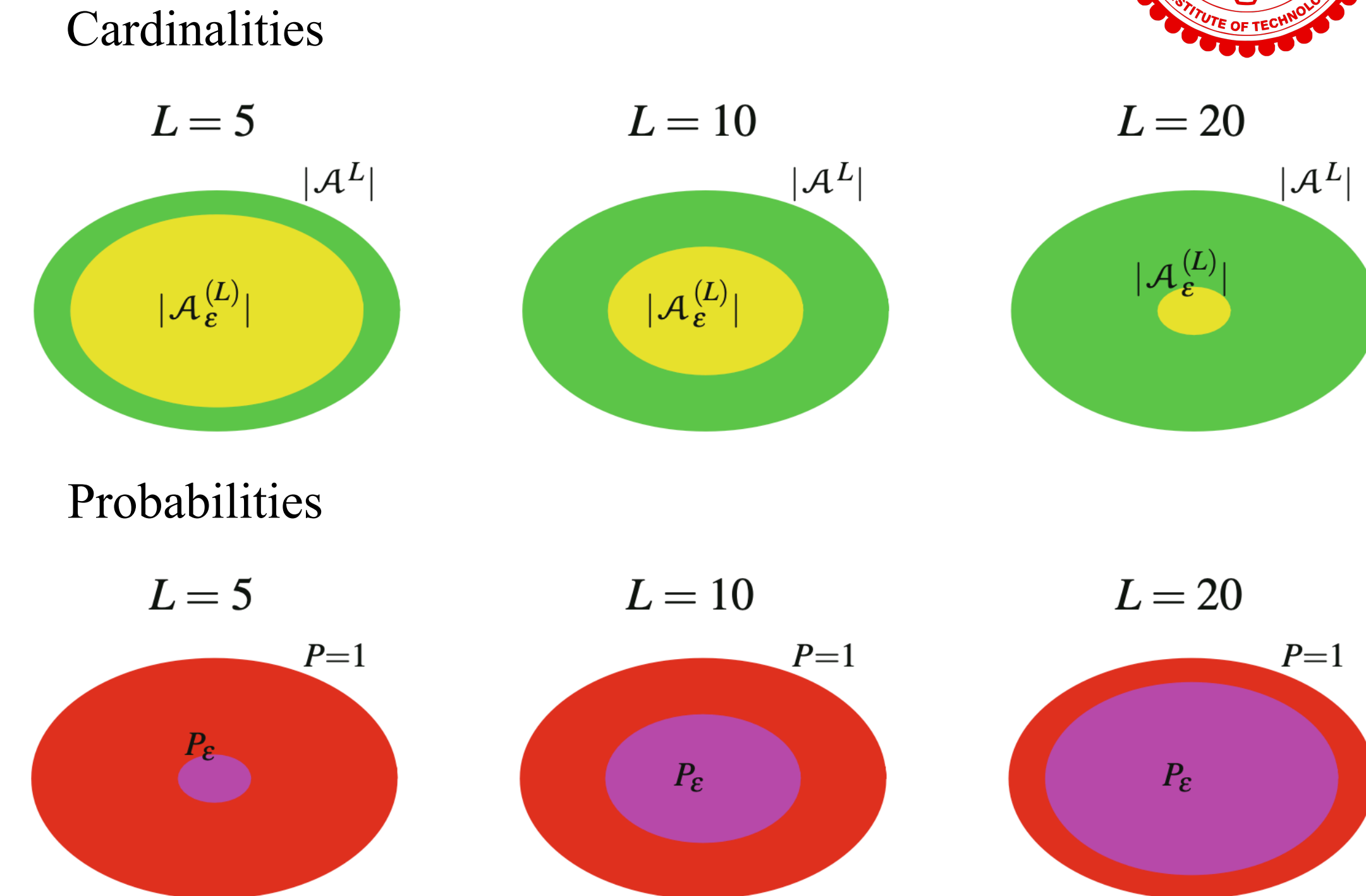


- **Asymptotic equipartition property (AEP):** Set of the possible sequences emitted by source is subdivided into two distinct classes, called *typical sequences* and *atypical sequences*.
- Example: IID binary source $P[A = 1] = p$ and $P[A = 0] = q = 1 - p$. For an L length sequence we expect to see in the sequence approximately Lp 1s, and Lq 0s (Typical Sequence)
- Probability of Typical sequence: $p(a^L) = p_A(a_1) \cdots p_A(a_L) \simeq p^{Lp} q^{Lq}$ and information: $i(a^L) \simeq -Lp \log p - Lq \log q = LH(A)$. Thus, $p(a^L) \simeq 2^{-LH(A)}$
- **Typical Sequences:** (1) are equally likely (uniformly distributed), and (2) their number is at most $2^{LH(A)}$
- **Shannon's Data Compression:** There are at most $N \simeq 2^{LH(A)}$ typical sequences, it is sufficient to create a codebook, or an index set, of cardinality N , where each codeword can be represented by a string of $\log N \simeq LH(A)$ bits, i.e., using $H(A)$ bits/symbol.
- **Definition:** A realization a^L is ε -typical if its sample entropy $\mathcal{J}(a^L) = -\frac{1}{L} \log p_A(a_1) \cdots p_A(a_L) = -\frac{1}{L} \sum_{n=1}^L \log p_A(a_n)$ is ε -close to the entropy $H(A)$, i.e., $H(A) - \varepsilon \leq \mathcal{J}(a^L) \leq H(A) + \varepsilon$.
- The ε -typical set $A_\varepsilon^{(L)}$ is the set of all ε -typical sequences s.t. $2^{-L(H(A)+\varepsilon)} \leq p(a_1) \cdots p(a_L) \leq 2^{-L(H(A)-\varepsilon)}$

Shannon's Compression Protocol

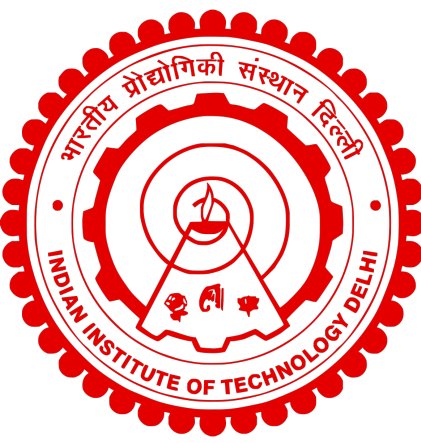


- Pictorially: ε -typical sequences.
- A compression or coding protocol $\mathcal{C}_L$ maps the sourceword $A^L = (A_1, \dots, A_L) \in \mathcal{A}^L$ into binary codeword $C^{LR} = (C_1, \dots, C_{LR})$ of LR bits.
- A decompression or decoding protocol $\mathcal{D}_L$ maps the LR bits back into a sequence $\tilde{A}^L = \mathcal{D}(C^{LR})$ of the original alphabet.
- Coding map: $\mathcal{C}_L : \mathcal{A}^L \rightarrow \{0,1\}^{LR}$ and decoding map: $\mathcal{D}_L : \{0,1\}^{LR} \rightarrow \mathcal{A}^L$
- Reliable protocol if: $\mathcal{D}_L(\mathcal{C}_L(A^L)) = A^L$ (*Asymptotically reliable* if this holds with probability one as $L \rightarrow \infty$)



The fraction of ε -typical sequences decreases with L .
The probability P_ε of ε -typical sequences increases with L

- The parameter R is called the *compression rate* and represents the number of bits/symbol used in the encoding.
- **Shannon's Noiseless Channel Coding Theorem:** Consider an IID source with entropy $H(A)$ and a compression rate of R bit/symbol. Then, if $R > H(A)$, there exists an asymptotically reliable compression protocol. Conversely, if $R < H(A)$ any compression protocol is not reliable.
- **Proof:** Application of properties of **Typical Sequences**



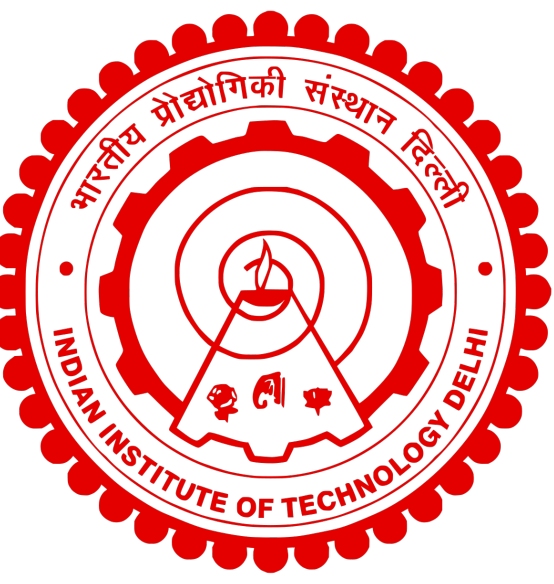
Shannon's AEP v/s Real Compression Algorithms

- Shannon's typical sequence (asymptotic equipartition property, AEP) is a theoretical foundation for compression, but it doesn't translate into a practical coder-decoder (codec).
- Requirements of “Typical-sequence compression”:
 - For a length- L source, identify the typical set (about 2^{LH} sequences)
 - Build a one-to-one mapping between each typical sequence and an index of $\approx LH$ bits.
 - Encoding: recognize which exact length- L sequence you have & output its index; Decoding: invert the map.
- Drawbacks of Shannon's AEP approach:
 - Need to enumerate/store an exponential number of sequences.
 - Need to know the true source distribution $P(x)$
 - Operate on very large blocks
 - Not practical: Requires exponential complexity and storage
- Practical Codecs (JPEG, PDF, etc) need low latency and streaming operation (byte-by-byte, block-by-block)
 - Uses the spatial redundancy, transforms and sparsity properties of Images and documents.
 - Practical codecs learn the probabilities online and handle non-stationarity of data

AEP gives the limit H (what's achievable)

Practical codecs give algorithms $\approx H$ (how to achieve it efficiently)

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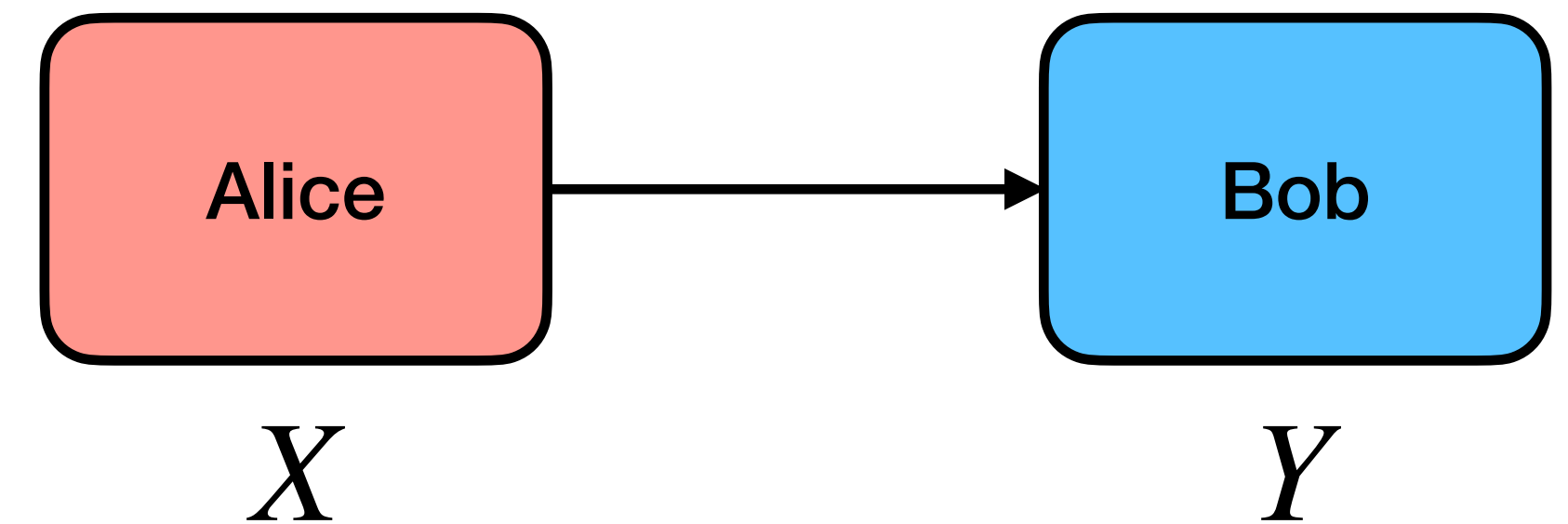


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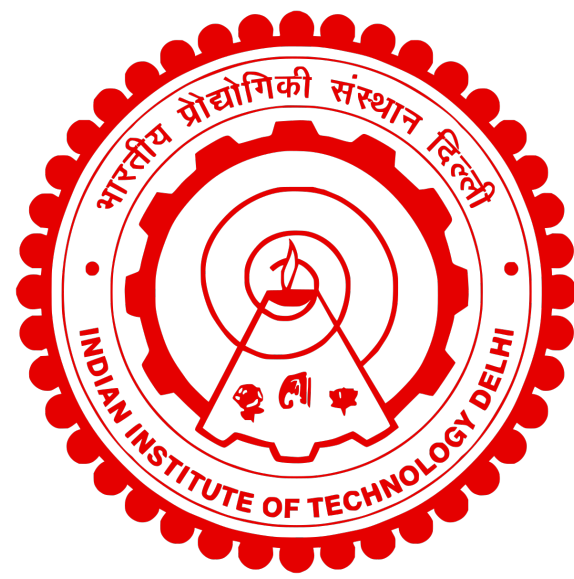
Conditional Entropy: Motivation



- Let X and Y be two random variables.
- $H(X)$ measures our uncertainty about X .
- If we observe Y , our uncertainty about X can reduce.
- This reduced uncertainty is quantified as the **Conditional Entropy: $H(X|Y)$** .
- Example Scenario:
 - Alice sends X to Bob over a noisy channel.
 - Bob receives Y , a noisy version of X .
 - Bob wants to estimate X given Y .



Conditional Entropy: Definition



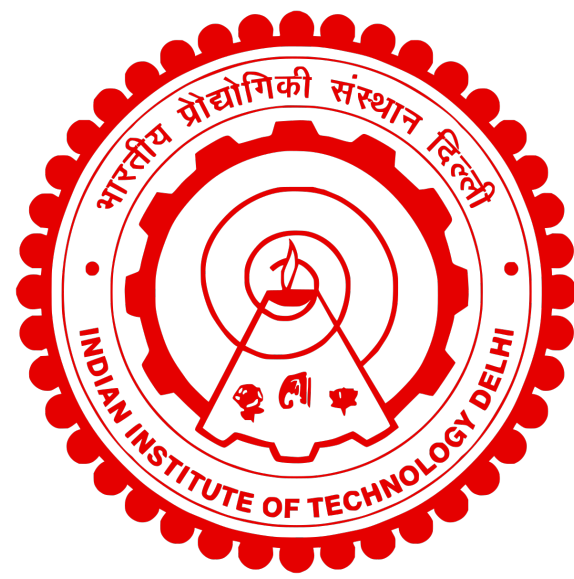
- **Given:** Joint distribution $p_{X,Y}(x, y)$ then $H(X | Y = y) = - \sum_x p_{X|Y}(x | y) \log p_{X|Y}(x | y)$

- Then, the **conditional entropy** is:

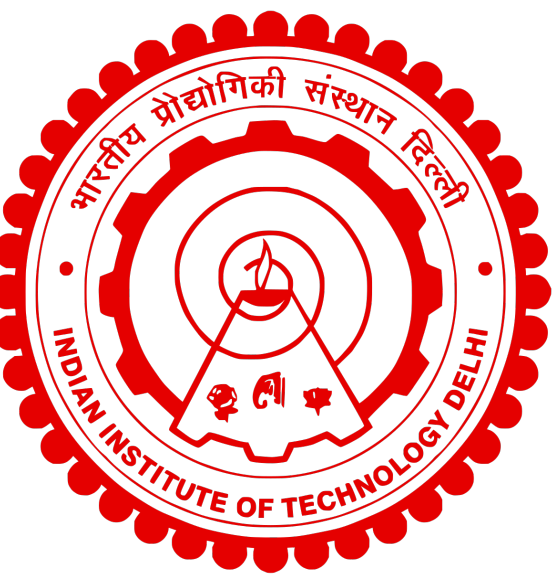
$$\begin{aligned} H(X | Y) &= \sum_y p_Y(y) H(X | Y = y) \\ &= - \sum_y p_Y(y) \sum_x p_{X|Y}(x | y) \log p_{X|Y}(x | y) \\ &= - \sum_{x,y} p_{X,Y}(x, y) \log p_{X|Y}(x | y) \end{aligned}$$

- **Special Case:** If $X \perp Y$, then $H(X | Y) = H(X)$

Conditioning Does Not Increase Entropy



- **Theorem:** For any two random variables X and Y , $H(X) \geq H(X|Y)$ with equality if and only if X and Y are independent.
- **Intuition:**
 - $H(X)$ represents the uncertainty about X before observing Y .
 - $H(X|Y)$ represents the remaining uncertainty after observing Y .
 - Gaining information (even noisy) about X via Y should not increase uncertainty.
- **Equality Condition:** $H(X) = H(X|Y)$ if and only if X and Y are independent, i.e., $p_{X,Y}(x,y) = p_X(x)p_Y(y)$
- **Interpretation:** Conditioning reduces or preserves entropy. It never increases it.



Joint Entropy

- Let X and Y be two random variables with joint distribution $p_{X,Y}(x, y)$.

- The **Joint Entropy** is: $H(X, Y) = - \sum_{x,y} p_{X,Y}(x, y) \log(p_{X,Y}(x, y))$

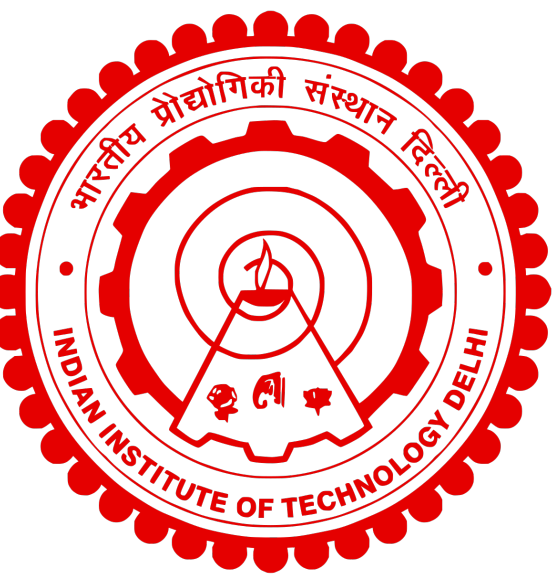
- **Chain Rule:** $H(X, Y) = H(X) + H(Y|X)$

- **Proof:** Using the identity $p_{X,Y}(x, y) = p_{Y|X}(y|x)p_X(x)$, we get:

$$\begin{aligned} H(X, Y) &= - \sum_{x,y} p_{X,Y}(x, y) \log \left(p_{Y|X}(y|x)p_X(x) \right) \\ &= - \sum_{x,y} p_{X,Y}(x, y) \log(p_X(x)) - \sum_{x,y} p_{X,Y}(x, y) \log(p_{Y|X}(y|x)) \\ &= H(X) + H(Y|X) \end{aligned}$$

- By symmetry: $H(X, Y) = H(Y) + H(X|Y)$
- Chain Rule shows how knowing one variable reduces uncertainty about the pair.

Mutual Information



- Let X and Y be discrete random variables with joint distribution $p_{X,Y}(x, y)$

- **Mutual Information** is defined as:

$$I(X : Y) = H(X) - H(X|Y) = H(Y) - H(Y|X)$$

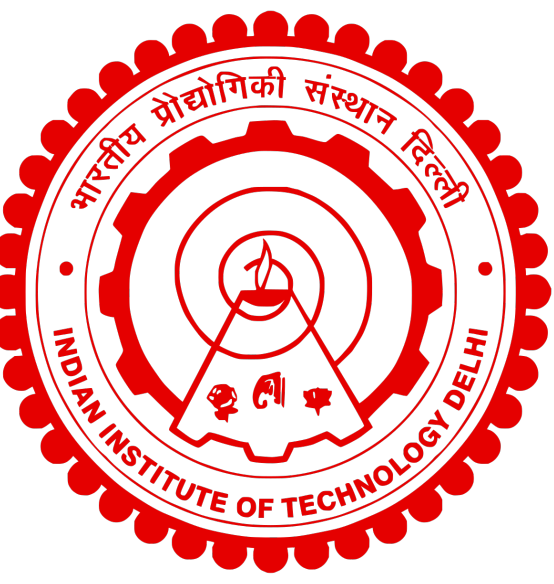
- **Interpretation:**

- $H(X)$: Uncertainty of X .
- $H(X|Y)$: Remaining uncertainty of X given Y .
- $I(X : Y)$: Amount of information X and Y share.

- **Properties:**

- If X and Y are independent, then: $H(X|Y) = H(X) \Rightarrow I(X : Y) = 0$
- **Theorem:** Mutual information is always non-negative: $I(X : Y) \geq 0$ with equality iff X and Y are independent.

Conditional Mutual Information



- Given a third variable Z , the **Conditional Mutual Information** is:

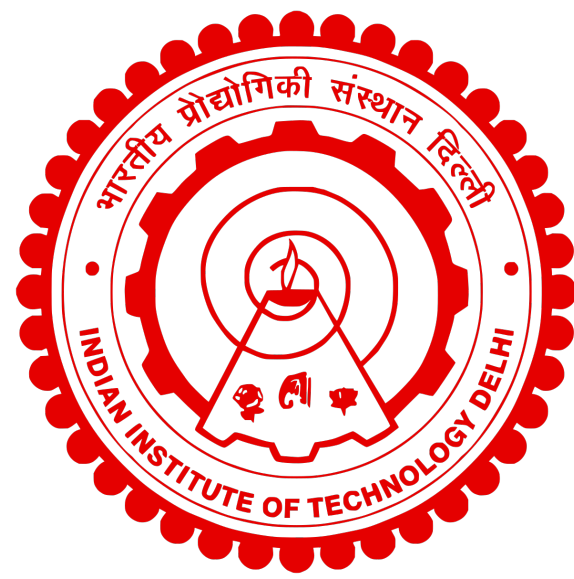
$$I(X : Y | Z) = H(Y | Z) - H(Y | X, Z)$$

- Alternative form: $I(X : Y | Z) = H(X | Z) - H(X | Y, Z) = H(X | Z) + H(Y | Z) - H(X, Y | Z)$
- Chain Rule for Mutual Information: Let $X_1, X_2, \dots, X_n$ and Y be random variables. Then:

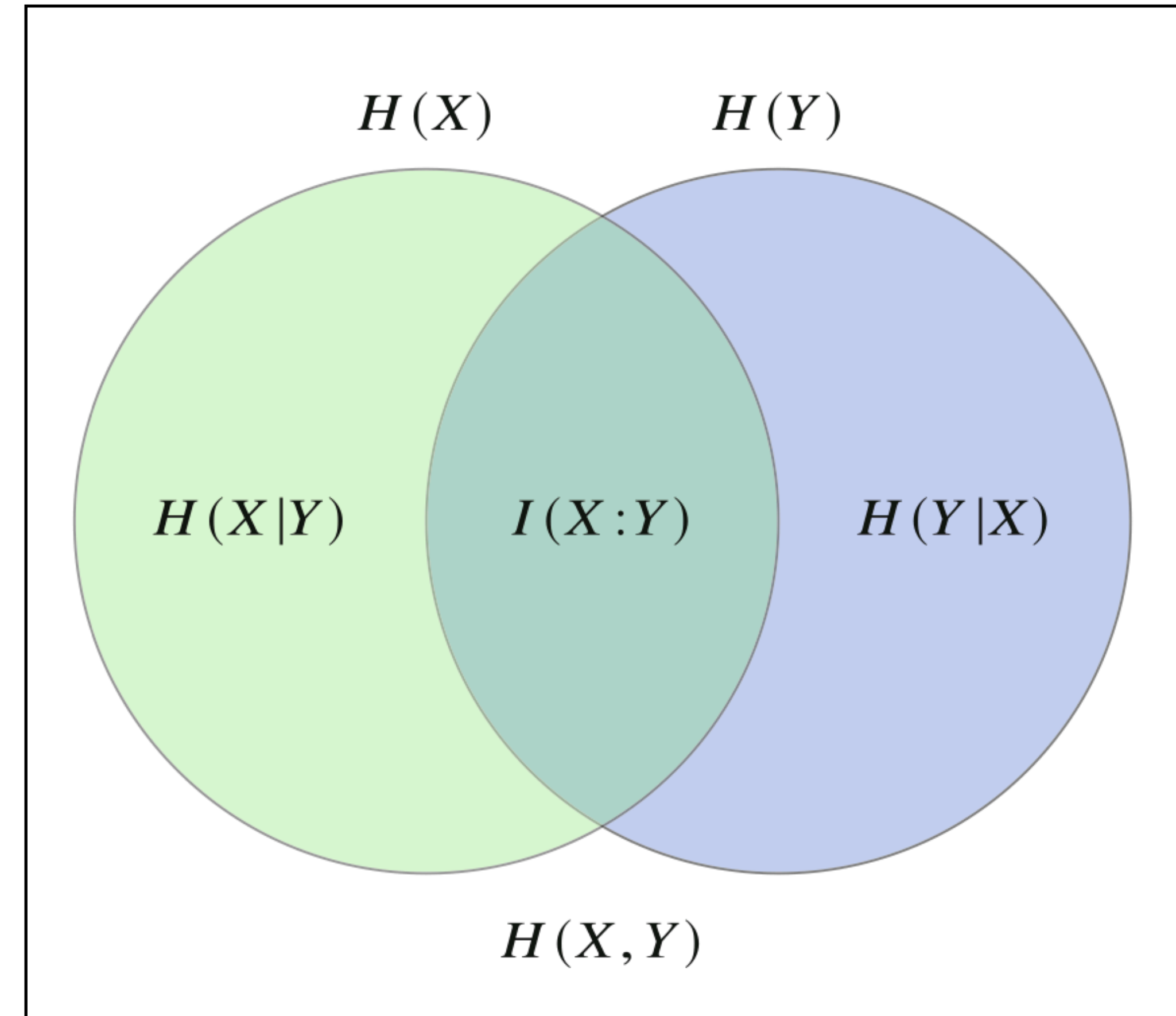
$$I(X_1, \dots, X_n : Y) = \sum_{i=1}^n I(X_i : Y | X_1, \dots, X_{i-1})$$

- This expresses mutual information as an accumulated quantity.

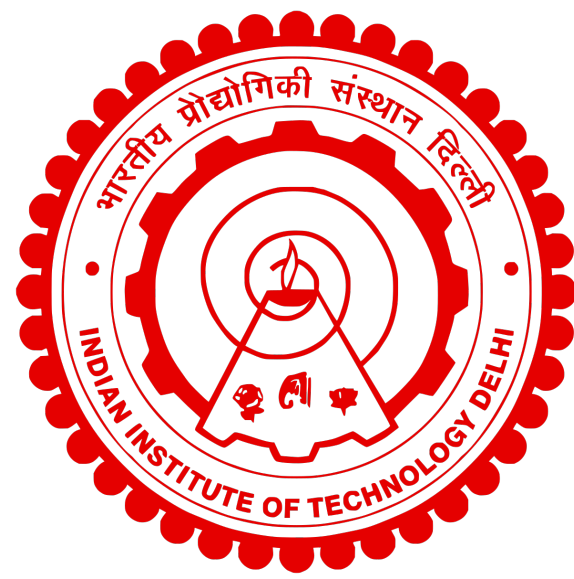
Relationship Between Entropies



- $H(X|Y)$: Uncertainty in X after knowing Y (green without overlap).
- $H(Y|X)$: Uncertainty in Y after knowing X (blue without overlap).
- $I(X : Y)$: Mutual information, the shared information between X and Y .
- $H(X, Y)$: Total uncertainty in the joint system (X, Y) , equal to the union area.
- Relation: $H(X, Y) = H(X|Y) + I(X : Y) + H(Y|X)$



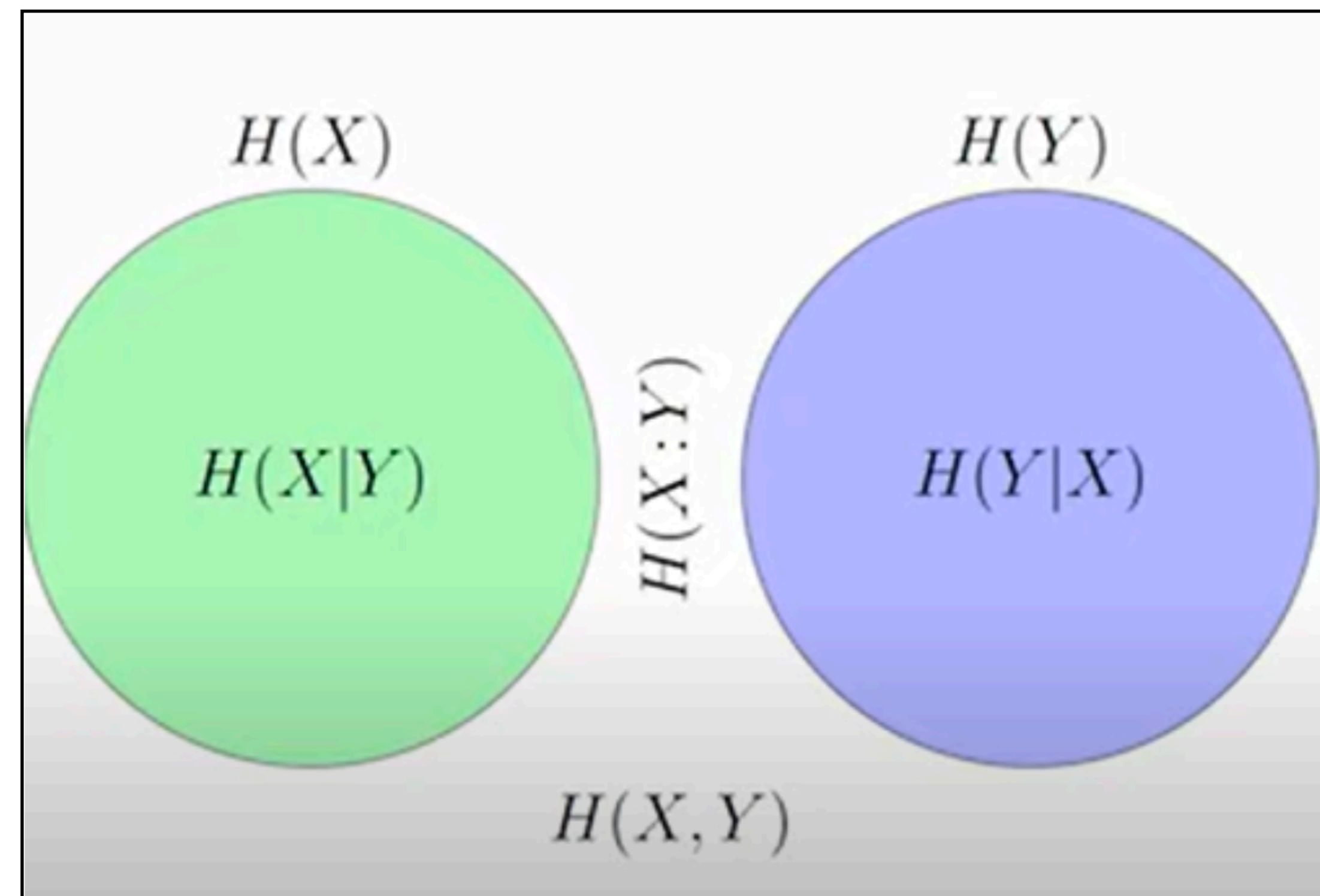
Mutual Information: Independence Case



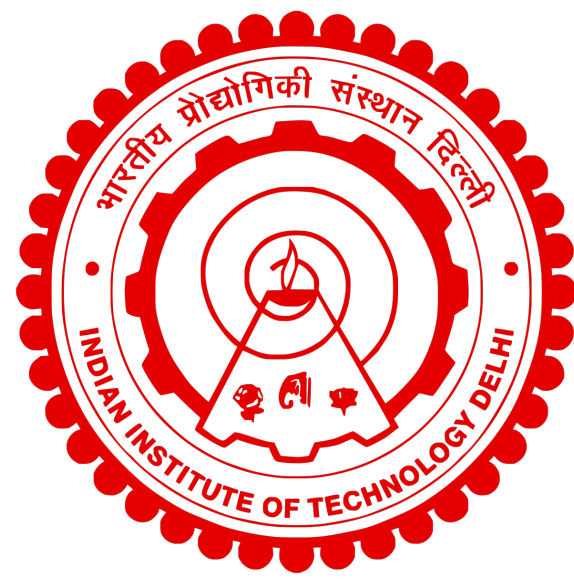
- The disjoint green and blue circles represent the entropies of X and Y respectively.
- Since there is no overlap, the mutual information is: $I(X : Y) = 0$
- This means X and Y are independent.
- Consequently:

$$H(X|Y) = H(X), \quad H(Y|X) = H(Y)$$

- The joint entropy simplifies to: $H(X, Y) = H(X) + H(Y)$



Relative Entropy



- **Definition:** Suppose $\mathcal{X}$ is a finite set and $p(x)$ and $q(x)$ are two probability distributions over the set $\mathcal{X}$. The relative entropy of $p(x)$ to $q(x)$ is:

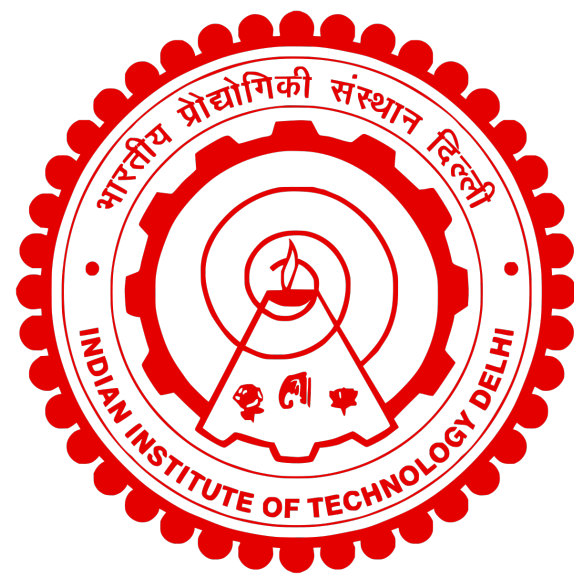
$$D(p(x) || q(x)) = \sum_x p(x) \log \left(\frac{p(x)}{q(x)} \right) = -H(X) - \sum_x p(x) \log(q(x))$$

- Where we define $-p(x) \log(0) = +\infty$ if $p(x) > 0$.
- **Theorem:** Given two probability distributions $p(x)$ and $q(x)$ over some alphabet $\mathcal{X}$, the relative entropy of $p(x)$ to $q(x)$ is non-negative:

$$D(p(x) || q(x)) \geq 0$$

- Equality holds if and only if $p(x) = q(x)$ for all $x \in \mathcal{X}$.

Example



p(x,y)	Y=0	Y=1
X=0	0.3	0.2
X=1	0.1	0.4

- Let X and Y be two binary random variables with the following joint probability distribution:

- Joint Entropy:** $H(X, Y) = - \sum_{x,y} p_{X,Y}(x, y) \log(p_{X,Y}(x, y))$

$$H(X, Y) = - [0.3 \log_2 0.3 + 0.2 \log_2 0.2 + 0.1 \log_2 0.1 + 0.4 \log_2 0.4] \approx 1.846 \text{ bits}$$

- Conditional Entropy:** $H(Y | X) = \sum_x P(x) H(Y | X = x)$

- Compute marginal: $P(X) : P(X = 0) = 0.3 + 0.2 = 0.5$,
 $P(X = 1) = 0.1 + 0.4 = 0.5$.

- Compute
 $H(Y | X = 0) = - [0.6 \log_2 0.6 + 0.4 \log_2 0.4] \approx 0.971$
(since $P(Y | X = 0) = [0.6, 0.4]$)

- Similarly, $H(Y | X = 1) \approx 0.721$

- Result:
 $H(Y | X) = 0.5 \times 0.971 + 0.5 \times 0.721 \approx 0.846 \text{ bits}$.

- Mutual Information:**
 $I(X : Y) = H(Y) - H(Y | X)$

- Compute:
 $H(Y) : P(Y = 0) = 0.3 + 0.1 = 0.4, P(Y = 1) = 0.6$,
 $H(Y) = - [0.4 \log_2 0.4 + 0.6 \log_2 0.6] \approx 0.971 \text{ bits}$

- Result: $I(X; Y) = 0.971 - 0.846 = 0.125 \text{ bits}$

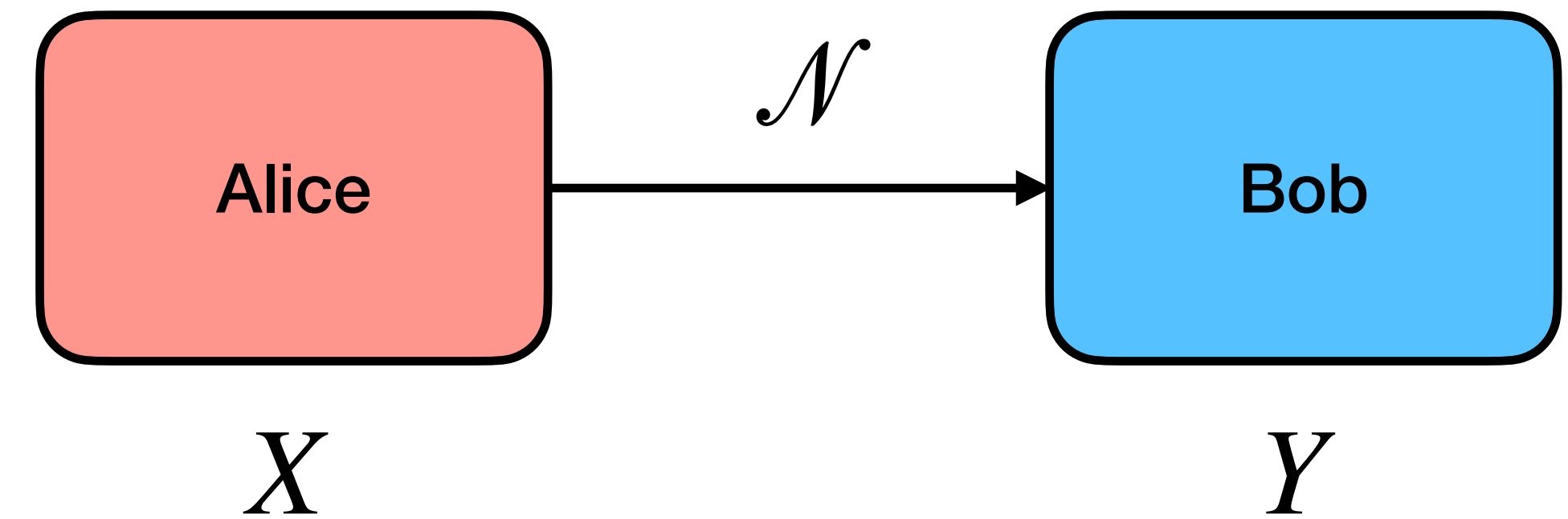
Shannon Channel Capacity



- Consider a noisy classical communication channel $\mathcal{N}$. Let:
 - X : input random variable (Alice's side)
 - Y : output random variable (Bob's side)
- The **capacity** of the channel is:

$$C(\mathcal{N}) = \max_{p_X(x)} I(X : Y)$$

- Optimized over all input distributions $p_X(x)$.
- Represents max information per channel use.



Shannon's Second Question: Channel Capacity



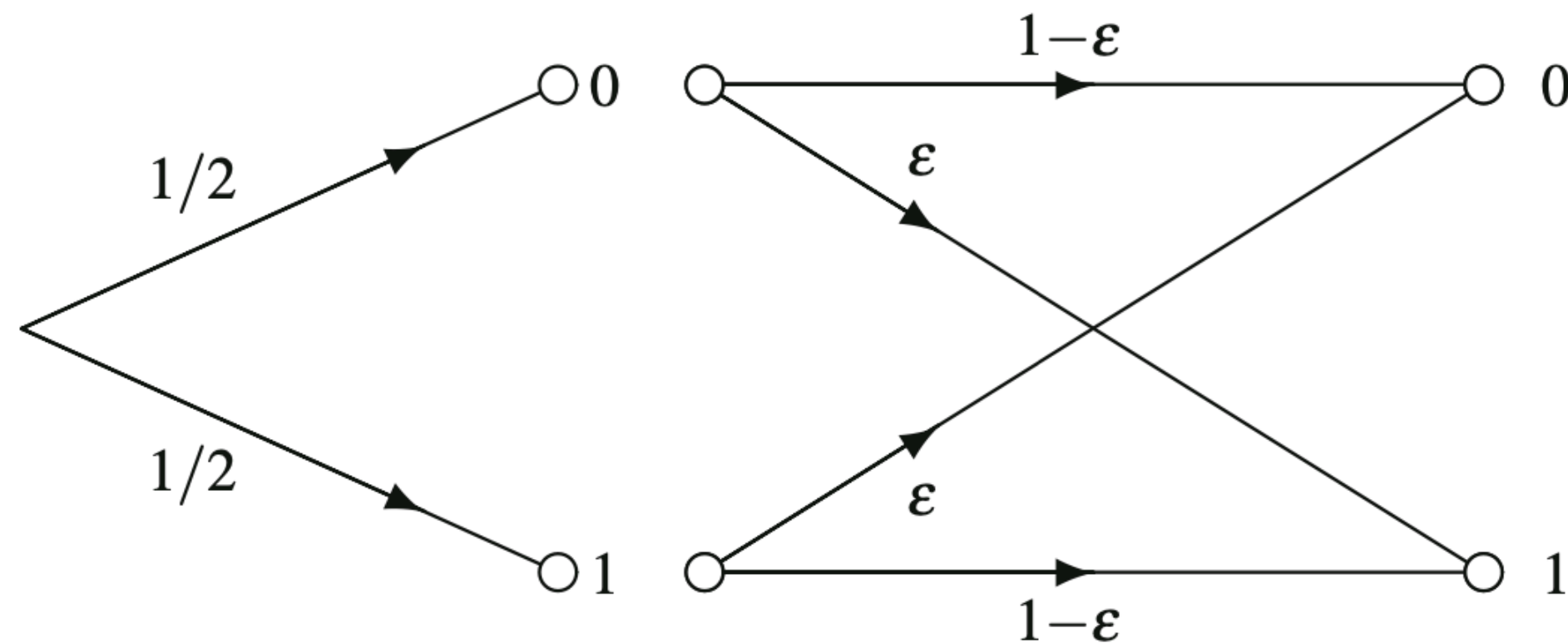
- **Q2:** *How much information can be reliably transmitted through a noisy communication channel?*
- **A2:** *According to **Shannon's Channel Coding Theorem**, the maximum reliable transmission rate is called the **Channel Capacity**:*

$$C(\mathcal{N}) = \max_{p_X(x)} I(X : Y)$$

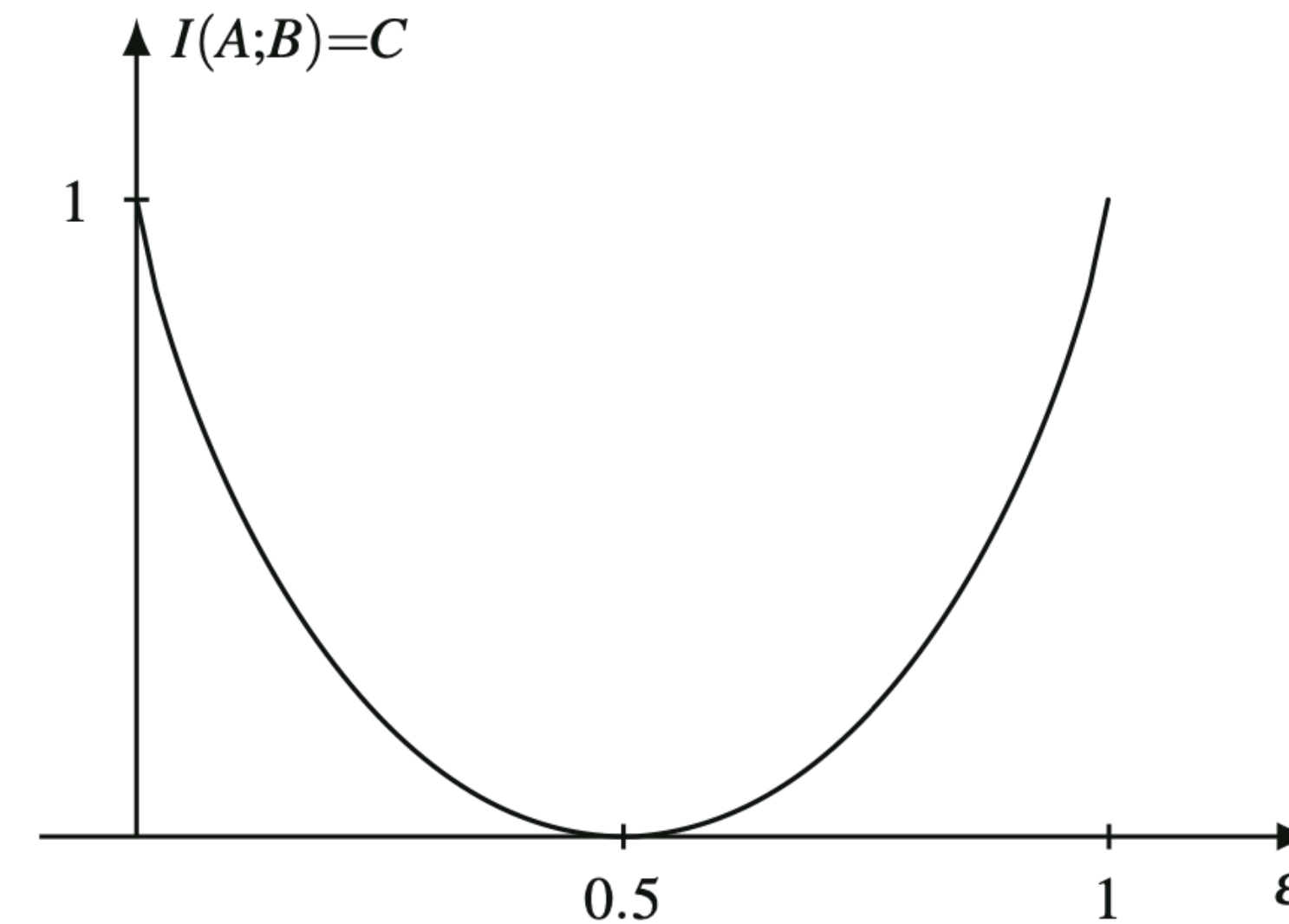
- **Explanation:**
 - Mutual Information $I(X : Y) = H(X) - H(X | Y)$ measures how much knowing Y reveals about X .
 - Capacity $C(\mathcal{N})$ represents the highest number of bits per channel use that can be transmitted with an arbitrarily low probability of error.
 - The maximization is over all input distributions $p_X(x)$, because different input strategies yield different efficiencies in using the channel.

Channel Capacity Example

- **Binary symmetric channel:** Cross transition probabilities: $p_c(0 | 1) = p_c(1 | 0) = \varepsilon$
- For equiprobable symbols $p_A(0) = p_A(1) = 1/2$, source entropy is $H(A) = 1$ bit/symbol
- Equivocation: $H(A | B) = -(1 - \varepsilon)\log(1 - \varepsilon) - \varepsilon \log \varepsilon$
- Mutual information: $I(A; B) = H(A) - H(A | B) = 1 + (1 - \varepsilon)\log(1 - \varepsilon) + \varepsilon \log \varepsilon$

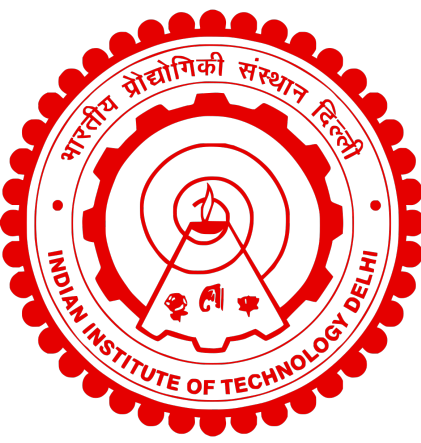


Binary symmetric channel



Channel Capacity = $I(A; B)$

Shannon's Noisy Channel Coding

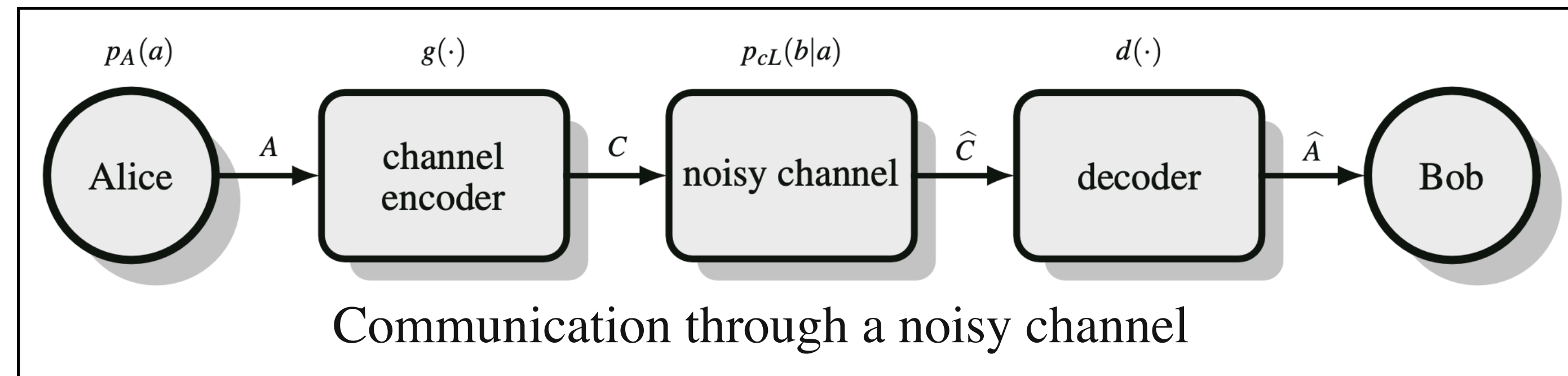


- Channel coding adds suitable redundancy in order to retrieve the original message with a small error probability (reliable communication)
- Coding in the Presence of Noise:

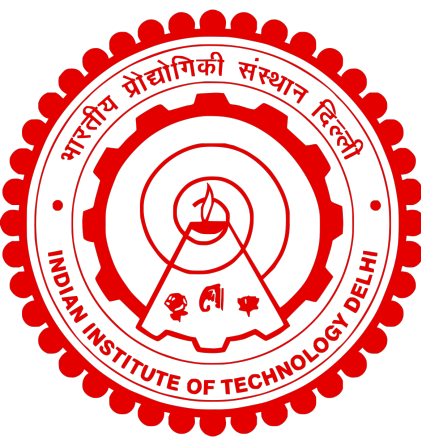
$A \in \mathcal{A} = \{1, \dots, K\}$ symbol (or message) produced by the source,
 $X^L = (X_1 X_2 \dots X_L) \in \mathcal{C}$ codeword sent to the channel,
 $Y^L = (Y_1 Y_2 \dots Y_L) \in \mathcal{Y}^L$ received codeword,
 $\hat{A} \in \mathcal{A}$ decoded symbol (or message).

- Channel Encoder: $x^L = g(a), a \in \mathcal{A}, x^L \in \mathcal{C}$
- L -extension of channel $(\mathcal{X}^L, \mathcal{Y}^L, p_{cL})$, where:

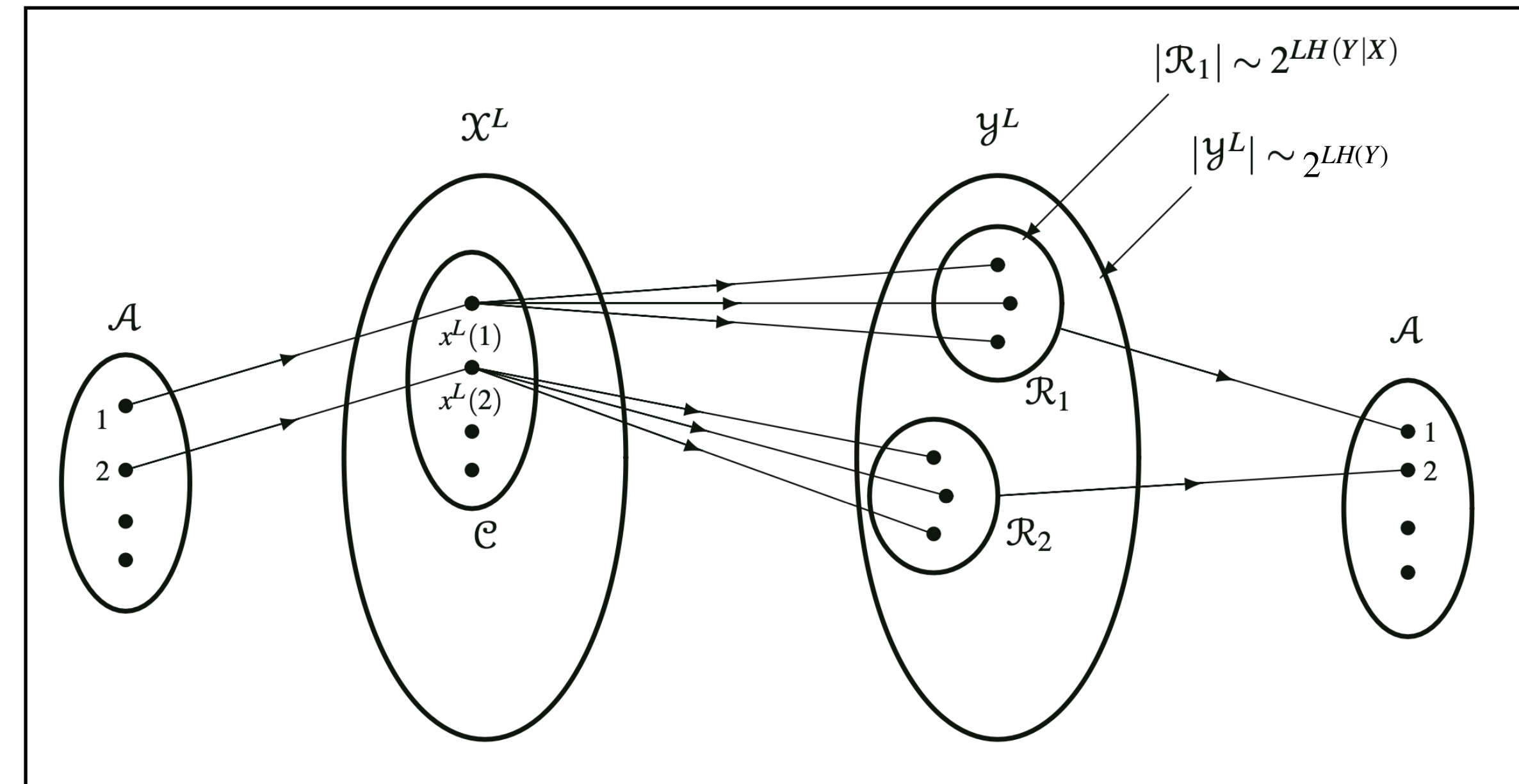
$$p_{cL}(y_1 y_2 \dots y_L | x_1 x_2 \dots x_L) = \prod_{n=1}^L p_c(y_n | x_n)$$
- Channel Decoder: $\mathcal{D} : \mathcal{Y}^L \rightarrow \mathcal{A}$, where
 $\hat{a} = \mathcal{D}(y^L), y^L \in \mathcal{Y}^L, \hat{a} \in \mathcal{A}$

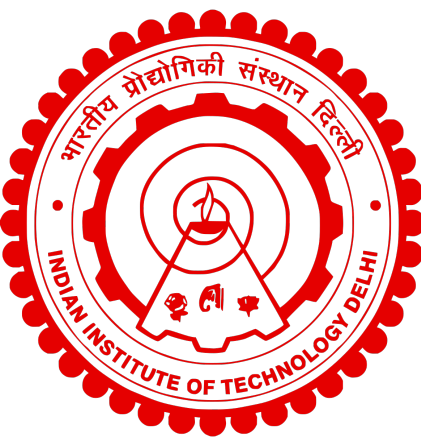


Shannon's Channel Coding Theorem



- **Theorem** (Shannon's second theorem): Given a noisy channel with capacity C and a source with K messages, provided that $\frac{1}{L} \log K < C$ there exists a coding and encoding system that guarantees an arbitrarily small error probability.
- Conversely, if $\frac{1}{L} \log K > C$ no encoding scheme guarantees arbitrarily small error probability.
- **Proof:** Based on random codeword: $x^L(a) = (x_1(a), x_2(a), \dots, x_L(a))$ with IID symbol and common distribution $p_X(x)$
- Highly likely that Y^L lies in the typical set with size $2^{LH(Y)}$
- If the input X^L is known then highly likely that Y^L lies in the conditional typical set of size $2^{LH(Y|X)}$
- How many codewords can we have such that the outputs are distinguishable?
- Sphere packing problem and intuitively: $\frac{2^{LH(Y)}}{2^{LH(Y|X)}} = 2^{LI(X;Y)}$
- The number of messages that can be faithfully transmitted is $|M| = 2^{LI(X;Y)}$ with rate equal to $\frac{\log_2 |M|}{L} = I(X; Y)$
- Optimize the rate by calculating $R = \max_{p_X(x)} I(X; Y)$



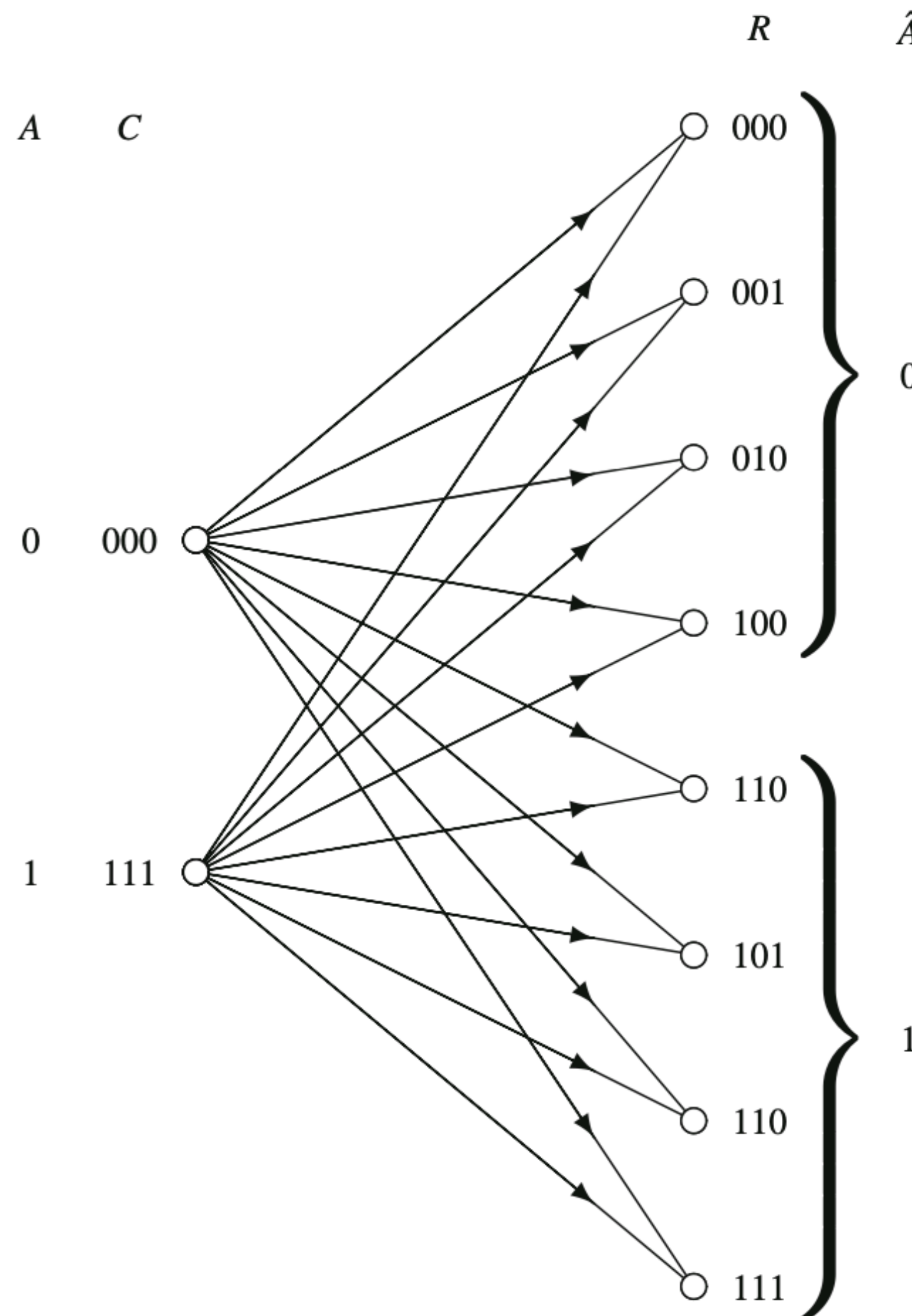


Repetition Code Example

- 3-1 repetition code

$\mathcal{A}$	$\mathcal{C}$	$\mathcal{R}$
0	000	000
		001
		010
		100
1	111	101
		110
		111

3-extension of of binary symmetric channel



- Prob. of error:

$$P_{e3} = P[\hat{A} \neq A] = \binom{3}{2} \epsilon^2(1 - \epsilon) + \epsilon^3$$

- Let $H(A) = 1$ bit/symbol (equiprobable symbols) and transmission rate: $f_c = 1000$ symbols/s with $\epsilon = 0.1$.

- Without encoding: Information rate:

$$H' = H(A)f_c = 1000 \text{ bit/s}$$

and error prob.: $P_e = \epsilon = 0.1$

- With 3-1 repetition code:

$$P_e := P[\hat{a} \neq a] = \binom{3}{2} \epsilon^2(1 - \epsilon) + \epsilon^3 = 0.028$$

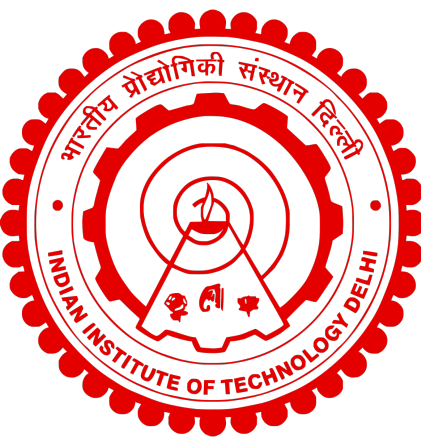
with information rate: $H' \approx 333$ bit/s ($f_s = f_c/3$).

- Increasing repetitions, Intuitively, $P_e \rightarrow 0$ provided that $H' = f_s H \rightarrow 0$.

- Shannon's counter intuitive conclusion: $P_e \rightarrow 0$ provided that $H' < C'$.

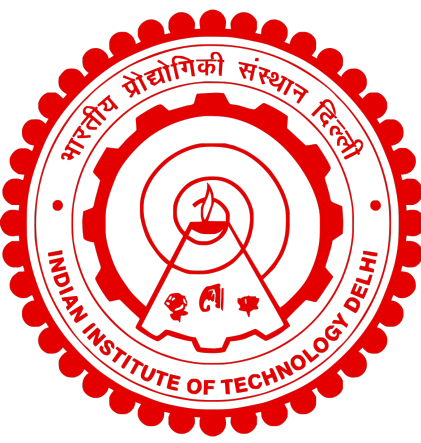
- Here $C' = f_c[1 + \epsilon \log \epsilon + (1 - \epsilon)\log(1 - \epsilon)] \approx 529$ bit/s

Summary - I



- Information in a message is related to its randomness or uncertainty
- Information source is a random variable A , specified by a probability distribution p_A
- Classical Shannon's Entropy (average information content): $H(A) := - \sum_a p_A(a) \log_2 p_A(a)$ (in bits)
- In practice, information sources are *redundant*, while ideal sources have uniform probability distribution.
- **Data compression/source coding:** Transform redundant source to uniform source to minimise bits/message.
- Classical **Shannon's source coding theorem:** For a sequence of i.i.d r.v. $(A_1, \dots, A_L)$, for $L \gg 1$, the minimum number of bits to represent such a sequence is given by the entropy of the sequence.

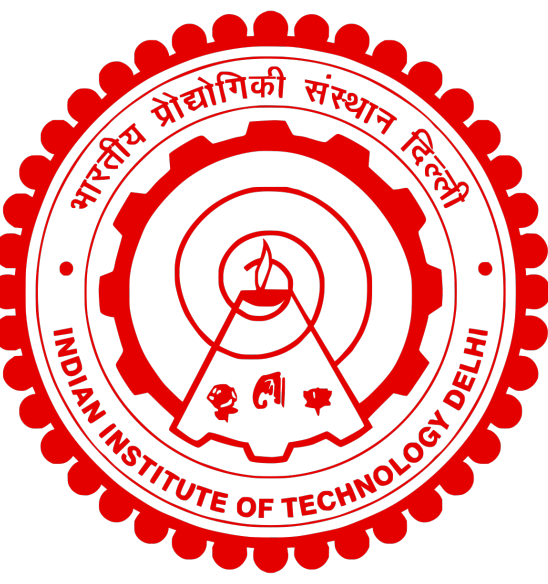
Summary - II



- **Channel coding** is used for reliable transmission of information over noisy channels.
- Classical coding: A source message of length L_0 is encoded into L -long codeword W from a prefixed codebook.
- The noisy channel produces the stochastic mapping: $W \rightarrow W'$ and Rate is defined as $R = L_0/L$
- **Shannon's second channel coding theorem:** For an appropriate choice of a codebook, there is a threshold called channel capacity C s.t. any $R < C$ is achievable with arbitrary small error probability.

Thank You

References



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